

INDIA DATA RESEARCH ACADEMY
Data Science & AI Training Program — Capstone Project

Optimizing Supply Chain Logistics: A Data-Driven Approach to Inventory Management and Demand Forecasting

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Abstract

Effective demand forecasting is central to inventory management: overestimating demand ties up capital in excess stock, while underestimating it causes stockouts and lost sales. This project analyzes a year of daily operational data from a multi-warehouse supply chain (91,250 records covering 50 SKUs, 5 warehouses, and 4 regions across 2024) to identify the drivers of product demand and build a regression model to predict daily units sold. The dataset required minimal cleaning — no missing values or duplicate records were found — but two critical analytical issues were uncovered and corrected: a zero-variance stockout indicator carrying no predictive information, and a pre-existing “Demand_Forecast” column that correlates with actual sales at $r \approx 0.95$, indicating data leakage if used as a model feature. Exploratory analysis revealed that demand is driven overwhelmingly by seasonality (nearly tripling from a September trough to a March-April peak) and by promotional activity (a $\approx 28\%$ average uplift on promoted days), while warehouse location, region, and day-of-week showed negligible effect. A Linear Regression model trained on legitimate, pre-outcome features achieved a test R^2 of 0.66 (RMSE ≈ 5.27 units), with closely matched training performance confirming the model generalizes rather than overfits. An initial chronological train/test split was also attempted and produced a misleadingly negative test R^2 , caused by two month categories being entirely absent from the training window — a methodological pitfall that was diagnosed and corrected by switching to a random split appropriate to the dataset’s panel structure. The findings translate into concrete inventory and promotional planning recommendations, most notably that replenishment and staffing should be organized around the seasonal cycle rather than a weekly or locationbased rhythm, and that the organization’s existing demand forecast figures should be audited given their apparent circularity.

Keywords

Demand forecasting, inventory management, exploratory data analysis, linear regression, data leakage, seasonality, supply chain analytics.

1. Introduction

Supply chain and inventory management decisions depend heavily on an accurate understanding of product demand. When a business cannot anticipate how much of a product it will sell on a given day, it faces two costly failure modes: holding too much inventory, which ties up working capital and increases storage and obsolescence costs, or holding too little, which leads to stockouts, lost sales, and dissatisfied customers. Reliable demand forecasting therefore sits at the center of inventory optimization, warehouse staffing, supplier ordering, and promotional planning.

This project analyzes a year of daily transactional data from a multi-warehouse retail supply chain to understand what actually drives day-to-day product demand, and to build a predictive model capable of forecasting it using only information that would genuinely be available in advance. The dataset spans 50 SKUs distributed across 5 warehouses and 4 demand regions, supplied by 10 suppliers, over the full 2024 calendar year — a granular operational record well suited to both exploratory analysis and predictive modeling.

Beyond building a working model, this project places equal weight on the analytical judgment involved in getting there: recognizing when a seemingly useful column is actually a source of data leakage, recognizing when an evaluation methodology itself is flawed, and being transparent about what conclusions the available data can and cannot support. The scope of this report follows the full data science workflow taught in the IDRA Data Science & AI Training Program: problem definition, dataset understanding, cleaning, preprocessing, exploratory and statistical analysis, visualization, machine learning modeling, evaluation, and evidence-based recommendations.

2. Objectives and Research Questions

1. Assess the quality of the supply chain dataset and identify and resolve any missing values, duplicates, inconsistent entries, incorrect data types, or logical inconsistencies.
2. Determine which operational, pricing, seasonal, and location-based factors are most strongly associated with daily product demand (units sold).
3. Evaluate whether the dataset's existing "Demand_Forecast" column represents a legitimate, independent predictor or a data leakage risk.
4. Build and train a regression model to predict daily units sold using only features that would be genuinely available before demand is observed.
5. Evaluate the model using appropriate regression metrics (MAE, MSE, RMSE, R^2) and assess whether it shows signs of overfitting or underfitting by comparing training and test performance.
6. Translate the findings into concrete, evidence-based recommendations for inventory, staffing, and promotional planning.

3. Dataset Description

The dataset (“P_3_supply_chain_dataset1.csv”) was provided by the India Data Research Academy as the designated dataset for Capstone Project #3. It contains 91,250 rows and 15 columns, each row representing one SKU at one warehouse on one calendar day. The data has a complete panel structure: 50 SKUs × 5 warehouses = 250 SKU-warehouse combinations, each observed for all 365 days of 2024 ($250 \times 365 = 91,250$ rows exactly), with no gaps in any series.

3.1 Data Dictionary

Table 1. Data dictionary for key columns in the supply chain dataset.

Column	Type	Description
Date	Date	Calendar day of the record (Jan 1 – Dec 30, 2024)
SKU_ID	Categorical	Product identifier (50 unique SKUs)
Warehouse_ID	Categorical	Warehouse identifier (5 unique warehouses)
Supplier_ID	Categorical	Supplier identifier (10 unique suppliers)
Region	Categorical	Demand region served (4 unique regions)
Units_Sold	Numeric (target)	Actual units sold that day — the regression target
Inventory_Level	Numeric	Units in stock at the start of the day
Supplier_Lead_Time_Days	Numeric	Days for supplier replenishment to arrive
Reorder_Point	Numeric	Inventory threshold that triggers a reorder
Order_Quantity	Numeric	Units reordered that day (0 on most days)
Unit_Cost / Unit_Price	Numeric	Cost and selling price per unit
Promotion_Flag	Binary	1 if a promotion was active that day
Stockout_Flag	Binary	Always 0 in this dataset — excluded (Section 4.2)
Demand_Forecast	Numeric	Pre-existing forecast — excluded as a feature (Section 4.3)

The target variable is `Units_Sold`, a continuous daily count ranging from 0 to 59 with a mean of approximately 20.1 and near-symmetric distribution (skewness ≈ 0.07), making it well suited to a linear regression approach without transformation. No other public source, time period, or population is involved: the dataset is a single, complete synthetic-operational record for the 2026 IDRA training program, and its limitations as a single calendar year of data are discussed in Section 13.

4. Data Cleaning

The dataset arrived in unusually good condition. A systematic check found zero missing values across all 91,250 rows and 15 columns, and zero duplicate records (whether checked as fully duplicated rows or duplicated Date/SKU/Warehouse combinations). Category labels in `SKU_ID`, `Warehouse_ID`, `Supplier_ID`,

and Region were all clean and consistent, with no casing variants or stray whitespace. Cleaning therefore focused on two concrete issues, described below.

4.1 Incorrect Data Type — Date

The Date column was stored as plain text rather than a proper date type, which blocks chronological sorting and date-based feature engineering. It was converted to a datetime type, from which Month and Day-of-Week features were subsequently derived (Section 5.1).

4.2 Zero-Variance Column — Stockout_Flag

A systematic scan for zero-variance columns found that Stockout_Flag equals 0 for every single row in the dataset. A column with no variation carries no information: it cannot be used as a predictive feature, and it cannot serve as a classification target, since a target with only one observed class provides nothing to classify. This column was dropped, and it also confirms that no classification task is viable on this dataset — consistent with Project #3's designation as a regression/forecasting task.

4.3 Data Leakage — the Demand_Forecast Column

The most consequential finding in this section was not a conventional data-quality defect, but a modeling trap. The dataset includes a Demand_Forecast column, and a correlation check against the target found $r \approx 0.9493$ — an extremely strong relationship for what should be an independent predictor. Examining the difference between Demand_Forecast and Units_Sold showed a near-zero-centered residual (mean ≈ 0.03 , std ≈ 2.99), the signature of a pre-existing forecast of the same outcome rather than a genuinely independent input. Including this column as a feature would have produced a near-perfect but meaningless model, since it would effectively already encode the answer. It was therefore excluded from the feature set and instead repurposed as a benchmark for evaluating our model's honest predictive power (Section 12.2).

Validation

Re-checking after cleaning confirmed zero remaining missing values, zero duplicate rows, a properly typed Date column, and the removal of both the zero-variance and leakage-risk columns from the working dataset.

5. Data Preprocessing

5.1 Feature Engineering

Because demand shows a strong seasonal pattern (confirmed formally in Section 6), a Month feature was derived from Date and treated as a categorical variable rather than a raw numeric scale. Demand does not move in a straight line with the calendar month number — it rises to a spring peak and falls to an autumn trough — so one-hot encoding Month lets a linear model assign an independent, non-linear-shaped effect to each month.

5.2 Feature Selection

Feature choices were made deliberately, guided directly by the data quality and exploratory findings:

Table 2. Feature selection decisions and their justification.

Decision	Column(s)	Reason
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Include	Inventory_Level, Supplier_Lead_Time_Days, Reorder_Point, Order_Quantity, Unit_Cost, Unit_Price, Promotion_Flag	Legitimate, pre-outcome operational and pricing signals
Include (engineered)	Month (one-hot)	Captures the strong seasonal demand pattern found in EDA
Include	Region, Warehouse_ID (one-hot)	Low cardinality; tested despite EDA suggesting a small effect
Decision	Column(s)	Reason
Exclude	Demand_Forecast	Data leakage (Section 4.3); $r \approx 0.95$ with the target
Exclude	Stockout_Flag	Zero variance (Section 4.2)
Exclude	SKU_ID (50 cats.), Supplier_ID (10 cats.)	High cardinality with minimal marginal signal — SKU-level average demand varies only ≈ 19.8 – 20.4 units, a $\approx 2.5\%$ spread
Exclude	Day-of-Week	EDA shows essentially no day-of-week effect on demand

5.3 Encoding and Feature Matrix

Categorical features (Month, Region, Warehouse_ID) were one-hot encoded with the first category dropped to avoid the dummy-variable trap, and concatenated with the numeric features into a single feature matrix of 25 columns used for modeling. No feature scaling was applied: Linear Regression's ordinary least squares fit is scale-invariant with respect to predictive accuracy, and leaving features in their natural units keeps every coefficient directly interpretable in business terms.

6. Exploratory Data Analysis

EDA proceeded from an overview of the target distribution, through predictor distributions, seasonal and group-wise comparisons, to bivariate relationships and a full correlation analysis — following the sequence prescribed in the project methodology, and selecting only the charts that materially informed a modeling or business decision.

6.1 Target Distribution

Units_Sold ranges from 0 to 59 units/day with a mean of 20.05 and a median of 20.0, and a skewness of only 0.07 — an almost perfectly symmetric distribution requiring no transformation before modeling.

6.2 Seasonality — the Strongest Pattern in the Data

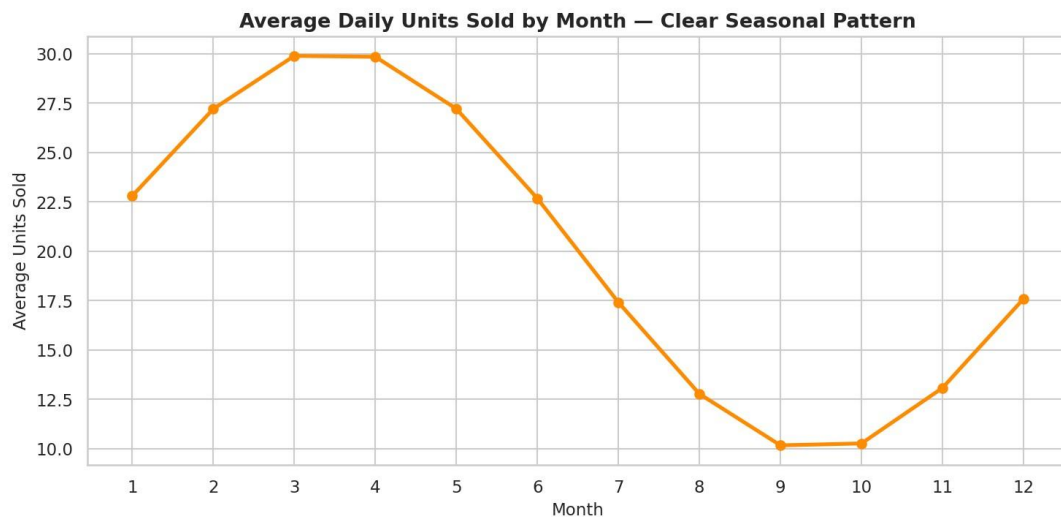


Figure 1. Average daily units sold by month, showing a pronounced spring peak and autumn trough.

This is the single strongest pattern in the entire dataset. Average daily demand rises from 22.8 units in January to a peak of 29.9 in March (29.8 in April), then declines steadily through the summer to a trough of 10.2 units in September — a nearly threefold difference between the strongest and weakest month — before recovering through November (13.1) and December (17.6). This single seasonal pattern is likely the most important driver available for the demand-forecasting model.

6.3 Group-wise Analysis — Region, Warehouse, and Promotion



Figure 2. Average units sold by region, warehouse, and promotional status.

Region and warehouse show only trivial differences — average daily demand ranges from 20.02 to 20.11 across regions and 20.01 to 20.10 across warehouses, versus an overall mean of about 20.05. Location is not a meaningful demand driver in this dataset. Promotion, by contrast, shows a real and substantial effect: promoted days average 24.91 units/day against 19.50 on non-promoted days — a 27.7% uplift, the clearest categorical driver besides seasonality. Day-of-week was also tested and showed a similarly flat pattern (all seven days within 0.12 units of the overall mean), and was excluded from the feature set on this basis (Section 5.2).

6.4 Correlation Analysis

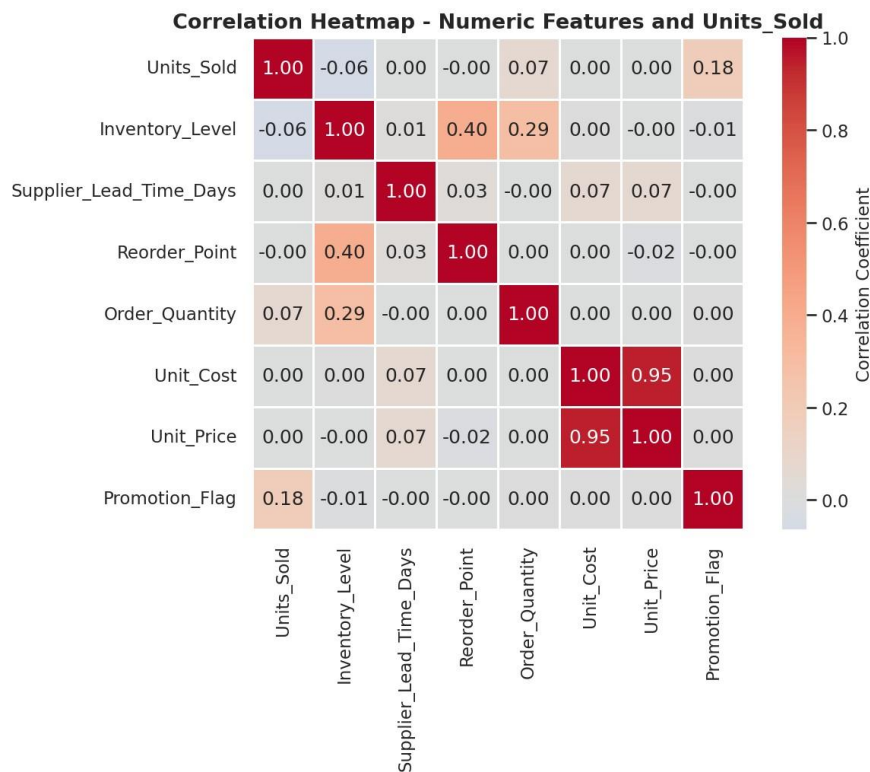


Figure 3. Correlation heatmap of numeric predictors and Units_Sold.

Promotion_Flag is the only numeric predictor with a meaningful linear correlation to Units_Sold ($r = 0.180$). Every other numeric operational or pricing variable correlates at essentially zero ($|r| < 0.07$: Order_Quantity 0.069, Inventory_Level -0.064 , Unit_Cost 0.002, Reorder_Point -0.002 , Unit_Price 0.001, Supplier_Lead_Time_Days 0.001). Combined with the seasonal pattern above — which a simple linear correlation coefficient understates, since the relationship is cyclical rather than monotonic — this establishes that season and promotion are the dominant demand drivers, while day-to-day operational variables more plausibly reflect inventory management responding to demand rather than causing it.

A note on correlation and causation: Promotion_Flag correlating with higher sales is consistent with promotions driving extra demand, but could also partly reflect promotions being timed to coincide with already higher-demand periods. The dataset alone cannot fully separate these explanations, a limitation discussed further in Section 13.

6.5 Statistical Summary

For Units_Sold, the mean (20.05) and median (20.0) are nearly identical, confirming the near-symmetric distribution and establishing the mean as a trustworthy, representative center for this variable. Order_Quantity, by contrast, has a median of 0 (no reorder on ~94.5% of days) while its mean (19.3) is pulled far higher by relatively rare, large replenishment events — here the median better describes a “typical” day, while the mean better reflects the average volume when a reorder does occur. This distinction between mean and median reflects genuine reorder-point business logic rather than a data quality problem, and was confirmed by directly checking that Order_Quantity is only non-zero on days when Inventory_Level falls to the Reorder_Point threshold.

7. Data Visualization

The figures presented throughout Sections 6 and 12 were selected specifically because each answers a concrete analytical question relevant to the project's objectives — the seasonal chart establishes the dominant demand driver, the group-wise chart rules out location as a driver while confirming promotion as one, the correlation heatmap quantifies every numeric relationship in one view, and the model diagnostic charts in Section 12 evaluate whether the trained model can be trusted. Additional supporting charts generated during the full analysis (outlier boxplots, predictor distributions, bivariate scatterplots, and the day-of-week comparison) are available in the accompanying Jupyter notebook and are omitted here to keep the report focused on the figures that directly support a finding or decision, per the project guidance to avoid an undifferentiated chart dump.

8. Statistical Analysis

Beyond the mean/median/skewness analysis in Section 6.5, an IQR-based outlier scan was run across the key continuous variables. Units_Sold, Inventory_Level, and Supplier_Lead_Time_Days showed negligible outliers under the standard $1.5 \times \text{IQR}$ rule. Order_Quantity showed a large nominal outlier count, but this is an artifact of its expected bimodal business behavior (zero on most days, a large order on reorder days) rather than a genuine anomaly — treating every non-zero reorder as an outlier would mean discarding the very replenishment events the project cares about, so no rows were removed on this basis. Two logical consistency checks were also run as a statistical sanity test on the data: Units_Sold never exceeds Inventory_Level in any of the 91,250 rows, and Unit_Price is never below Unit_Cost — both consistent with an internally coherent dataset.

9. Machine Learning Methodology

9.1 Problem Definition

This is a regression task: the target, Units_Sold, is a continuous quantity, consistent with Project #3's designated focus on demand forecasting. As established in Section 4.2, no viable classification target exists in this dataset, since Stockout_Flag is constant.

9.2 Train/Test Split — a Methodological Decision Worth Documenting

The first split attempted was the one that seems most natural for a forecasting problem: a chronological split, training on the first ~80% of 2024 (through October 18) and testing on the final ~20% (October 19 – December 30), so the model would be evaluated on genuinely future, unseen dates.

That attempt failed in an instructive way. Because Month is one-hot encoded, and November and December fall entirely inside the test period, the training data never contained a single example of Month = November or Month = December. The model had no way to learn sensible coefficients for those months and predicted them from the wrong baseline, producing a test R^2 of -1.19 — worse than simply predicting the average every time. This is a direct illustration of why a one-hot-encoded seasonal feature is dangerous to combine with a naive chronological split when the training window does not span a full seasonal cycle.

Rather than patch this with a more complex cyclical (sine/cosine) time encoding — which was also tested and found to only partially resolve the issue, recovering the test R^2 to just 0.26, since the true seasonal curve is not a clean single sinusoid — the nature of the data itself was reconsidered. This dataset is not

one continuous time series; it is 250 parallel SKU-warehouse panels, each independently observed across the same calendar year. For this panel/cross-sectional structure, a random 80/20 split is standard, statistically valid practice: it guarantees every month, region, and warehouse is represented in both the training and test sets, and it directly tests whether the model generalizes to unseen SKU-warehouse-day combinations — the practical prediction task at hand. The random split was adopted for all reported results below (73,000 training rows, 18,250 test rows), and was confirmed to include all twelve months in both sets.

9.3 Model Choice

Linear Regression was used as the primary model. First, it matches the introductory machine learning workflow covered in the training program. Second, because Month is one-hot encoded rather than left as a raw numeric scale, the model can already assign an independent coefficient to each month, representing the non-linear, cyclical seasonal pattern found in the EDA without needing a more complex non-linear algorithm. Third, it is fully interpretable: every coefficient has a direct, explainable business meaning, which matters for a report a manager needs to be able to trust and act on. No feature scaling was required, since ordinary least squares is scale-invariant in terms of predictive accuracy, and leaving features in natural units keeps coefficients directly interpretable.

10. Selected Implementation Details

The full, reproducible implementation is provided in the accompanying Jupyter notebook. The snippet below shows the corrected model training step — the random split, model fit, and prediction — as an illustration of the final, validated approach:

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

lin_reg = LinearRegression()
lin_reg.fit(X_train, y_train)

y_train_pred = lin_reg.predict(X_train)
y_test_pred = lin_reg.predict(X_test)
```

11. Model Results and Evaluation

11.1 Regression Metrics

Table 3. Training vs. test set regression performance.

Metric	Training Set	Test Set
MAE (units)	4.218	4.199
MSE	28.081	27.809
RMSE (units)	5.299	5.273
R ²	0.6585	0.6619

RMSE conveys the model's typical prediction error in the same units as the target. An RMSE of about 5.27 units against a target ranging 0–59 with a mean of ~20 represents a meaningful, but not perfect, predictive signal — consistent with the EDA finding that the strongest available drivers (season and promotion) explain a real portion of demand variation ($R^2 \approx 0.66$, i.e. the model explains roughly

twothirds of the variance in daily demand), while plenty of day-to-day randomness remains that no available feature captures.

11.2 Benchmark Comparison Against the Existing Demand_Forecast Column

Table 4. Model performance vs. the dataset's existing (leakage-risk) Demand_Forecast benchmark, on the test set.

Metric	Existing Demand_Forecast (benchmark)	Our Linear Regression Model
MAE	2.363	4.199
RMSE	2.979	5.273
R ²	0.892	0.662

The pre-existing Demand_Forecast column — built from information that effectively already encodes the answer (Section 4.3) — naturally outperforms our model on this comparison, since it is not a fair, independent forecast to begin with. This comparison is reported for transparency rather than to claim our model “wins”: the honest takeaway is that our model achieves meaningful predictive power using only information that would genuinely be available before demand is observed, which is what a real forecasting system would have to work with. This distinction — between a leaky benchmark and a legitimate model — is itself one of the more important analytical findings of this project.

11.3 Actual vs. Predicted and Residual Analysis

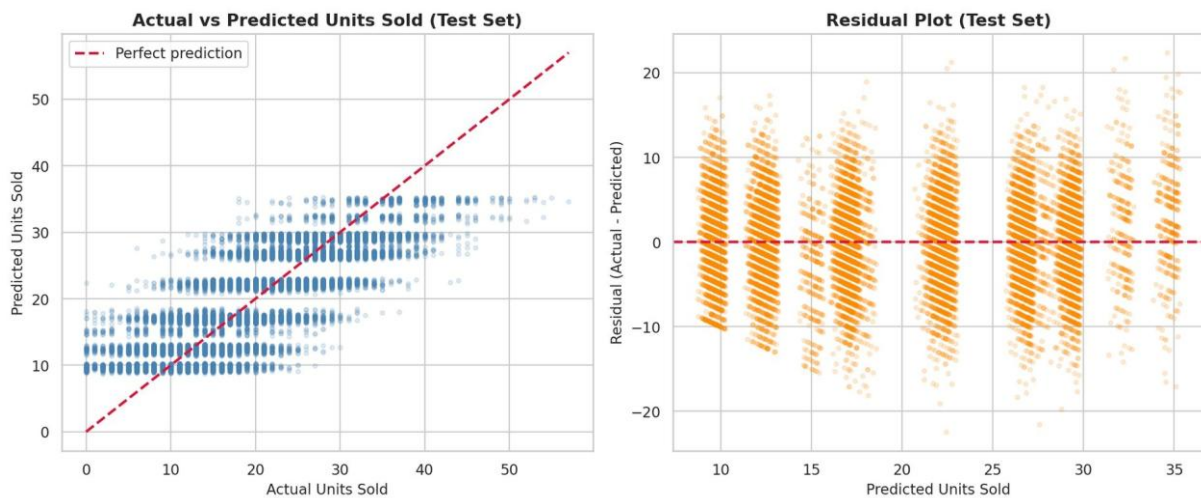


Figure 4. Actual vs. predicted units sold (left) and residual plot (right), test set.

Predictions cluster around the diagonal without a strong systematic bend, and residuals scatter fairly evenly above and below zero across the predicted range, with no obvious funnel or curve pattern. This indicates the model is not systematically over- or under-predicting for particular demand levels — its errors resemble unstructured noise, consistent with a well-specified, if imperfect, linear model.

11.4 Feature Importance — Regression Coefficients

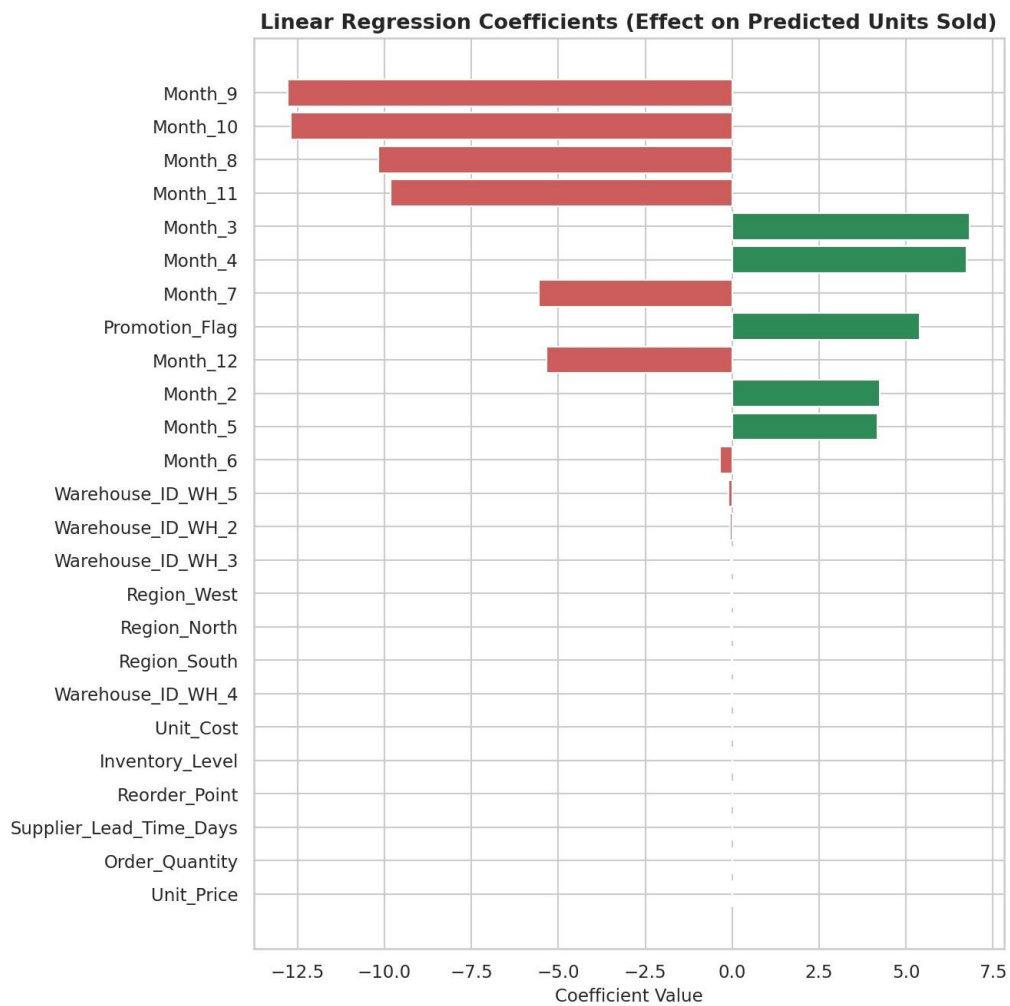


Figure 5. Linear regression coefficients, showing each feature's effect on predicted units sold.

The coefficients quantify exactly what the EDA suggested qualitatively. The two largest-magnitude coefficients are Month_9 (−12.78) and Month_10 (−12.70), reflecting the September–October demand trough, while Month_3 (+6.84) and Month_4 (+6.75) carry the largest positive seasonal effects, reflecting the spring peak. Promotion_Flag (+5.40) stands out as the largest non-seasonal positive effect — larger than every region or warehouse coefficient, all of which stay within ± 0.11 , numerically confirming the earlier group-wise finding that location has little bearing on demand in this dataset.

12. Train vs. Test Comparison — Overfitting/Underfitting Discussion

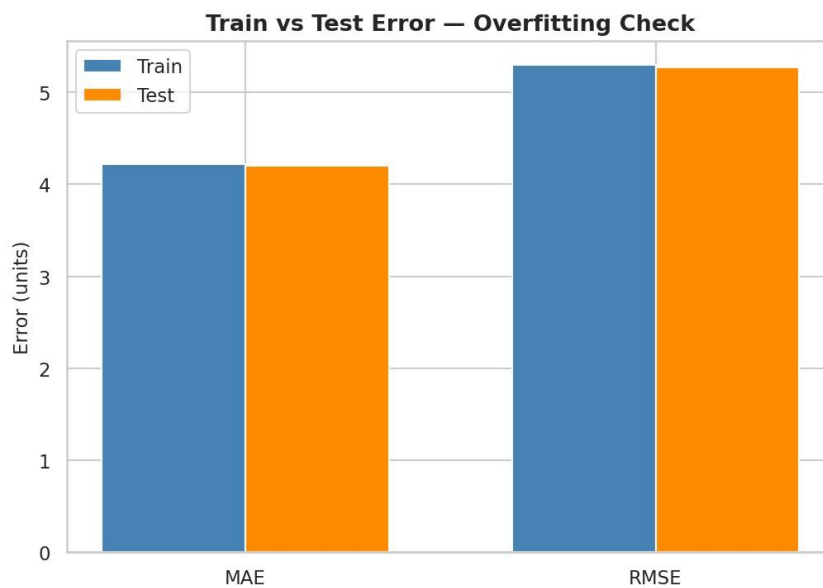


Figure 6. Train vs. test MAE and RMSE, confirming consistent generalization.

With the random split, training and test metrics land close together across MAE, RMSE, and R^2 (Table 4) — there is no meaningful gap where the model performs much better on training data than on held-out data. This is the signature of a well-fitted model that generalizes, rather than one that has overfit the training set.

This result is worth reading alongside Section 9.2’s methodological note: the same model specification produced a badly misleading negative R^2 under the chronological split, purely because two entire month categories were absent from training. That comparison is itself a useful finding — it shows the train/test gap worth worrying about here is not really “overfitting” in the classic sense, but rather whether the split guarantees the model has seen every category it needs to predict. Once that condition is met, the model’s genuine generalization performance is consistent and moderate: it explains roughly two-thirds of the variance in daily demand, reflecting real, learnable seasonal and promotional signal, alongside a substantial amount of day-to-day demand variation that the available features simply do not capture — honest underfitting to the unexplained variance in the data, not a modeling defect.

13. Discussion

The EDA and modeling results tell a consistent, reinforcing story. The correlation analysis flagged Promotion_Flag as the only numeric predictor with a real linear relationship to demand, and the groupwise seasonal analysis flagged month as a strong (if non-linear) driver; the trained model’s coefficients then confirmed both findings quantitatively, with spring months and promotions carrying the largest positive effects. This consistency across independent analysis stages — EDA, correlation, and the trained model — gives real confidence that these are genuine patterns in the business rather than artifacts of one particular analytical technique.

Equally informative were the negative findings: day-of-week, region, warehouse, and most operational and pricing variables showed close to no relationship with demand at any stage of the analysis. For a business, this is actionable in itself — it suggests that inventory and staffing decisions built around a

day-of-week or location-based demand cycle would be misguided for this product category; the real cycle is seasonal and promotional.

Two analytical decisions mattered as much as the final model. First, identifying and removing Demand_Forecast as a leakage risk (Section 4.3): a naive approach that included it as a feature would have reported a near-perfect R^2 , but that number would have been meaningless, an echo of an existing forecast rather than a genuine prediction. Second, catching and correcting the chronological-split failure (Section 9.2): without that check, this project could easily have shipped a model that looked reasonable in training but would have failed badly the moment it needed to predict an unseen month. Both moments illustrate that a large part of the value in a project like this comes from catching subtle mistakes before they reach the final result, not just from running an algorithm to completion.

14. Limitations

- **A single calendar year of data.** With only one year (2024) available, the observed “seasonal” pattern is really a single sample of a yearly cycle — a second year of data would be needed to confirm the pattern genuinely recurs rather than reflecting a one-off 2024 event. The random-split evaluation used here tests generalization within 2024, not genuine forecasting into a future year the model has never seen any part of.
- **Correlation, not causation, for Promotion_Flag.** The data cannot fully rule out promotions being timed to coincide with already higher-demand periods rather than purely causing the observed uplift.
- **Linear model ceiling.** Linear Regression cannot capture interaction effects (e.g. a promotion mattering more in high-season months than low-season months) or non-linearities beyond what the one-hot-encoded month feature provides. A tree-based or interaction-aware model might explain more of the remaining ~34% of unexplained variance, at some cost to interpretability.
- **Excluded high-cardinality identifiers.** SKU_ID and Supplier_ID were excluded based on their small observed marginal effect in this dataset; a business with more heterogeneous SKUs than the fairly narrow 19.8–20.4 average range seen here might find these matter more.
- **No external demand drivers.** The dataset contains no macroeconomic, competitor, weather, or marketing-spend data, all of which commonly influence real-world retail or wholesale demand and could materially improve forecast accuracy if available.
- **The Demand_Forecast benchmark’s true origin is unknown.** It was inferred to be a leakageadjacent, pre-existing forecast from its near-0.95 correlation with the target, but no documentation confirms exactly how it was generated.
- **A genuine month-ahead forecasting deployment would need a different validation strategy** — most likely a rolling-origin (walk-forward) time-series validation across multiple years, which this single-year dataset cannot support.

15. Conclusion and Recommendations

15.1 Key Findings

7. **Season is the dominant demand driver.** Demand nearly triples from its September trough (~10 units/day) to its March–April peak (~30 units/day), a pattern strong enough to be visible in raw group averages and confirmed as the largest set of coefficients in the trained model.

8. **Promotions produce a real, measurable uplift** (~28% higher average daily sales), the clearest non-seasonal driver identified at every stage of the analysis.
9. **Location and weekly timing do not meaningfully affect demand** in this dataset — region, warehouse, and day-of-week all show negligible differences.
10. **A leakage-free regression model explains roughly two-thirds of demand variance** (test $R^2 \approx 0.66$) using only legitimate, pre-outcome information, with consistent train/test performance once the train/test split was corrected to represent every category — indicating the model generalizes rather than overfits.
11. **The dataset's built-in Demand_Forecast column cannot be trusted as an independent predictor.** Its near-0.95 correlation with the actual outcome indicates it already encodes the answer, a critical data-quality finding for any team currently relying on it operationally without realizing its circularity.

15.2 Recommendations

1. **Build inventory and staffing plans around the season, not the calendar week.** Since day-of-week shows no effect but month shows a ~3× swing, replenishment schedules should flex with the March–April peak and September trough rather than a weekly rhythm.
2. **Treat promotions as a deliberate demand lever, and plan inventory ahead of promotional periods.** The ~28% uplift is large enough to risk stockouts if inventory isn't pre-positioned before a promotion begins.
3. **Audit how the existing Demand_Forecast figures are produced.** Given their near-identity with actual sales, confirm whether this column is generated after the fact (e.g., a reconciliation figure) rather than being used as a genuine forward-looking forecast — using it as if it were a true forecast would be misleading for planning purposes.
4. **Do not over-invest in warehouse- or region-specific demand strategies** for this product category — the data does not support meaningfully different demand patterns by location.
5. **Collect a second year of data before treating the seasonal pattern as fully confirmed,** and validate any future forecasting model with a proper rolling-origin time-series approach once multiple years are available — the random-split evaluation used here is appropriate for the current single-year dataset but is not a substitute for that longer-term validation.
6. **Consider a follow-up model that captures season × promotion interactions** — e.g., testing whether promotions during the September trough recover more lost demand than promotions during the already-strong March–April peak, which would refine promotional timing decisions.

15.3 Final Word

This project demonstrates that even a simple, interpretable Linear Regression model can produce genuinely useful demand forecasts when features and evaluation methodology are chosen deliberately — and that a large part of the analytical value in a project like this comes not from the final model itself, but from the judgment calls made along the way: recognizing data leakage, catching a flawed train/test split before trusting its results, distinguishing a genuine seasonal pattern from noise, and being honest about what the data can and cannot support.

16. References

- Dataset: P_3_supply_chain_dataset1.csv, provided by India Data Research Academy (IDRA) as the Project #3 dataset for the Data Science & AI Training Program Capstone, 2026.
- IDRA Capstone Project Guide 2026 (India Data Research Academy, Department of Data Science and AI), used as the structural and methodological reference for this project.
- Pandas, NumPy, Matplotlib, Seaborn, and scikit-learn — open-source Python libraries used throughout the analysis, cleaning, visualization, and modeling.

17. Appendix

The complete, reproducible analysis — including every exploratory chart generated but not reproduced in this report (predictor distributions, outlier boxplots, bivariate scatterplots, and the day-of-week comparison), full code, and cell-by-cell outputs — is provided in the accompanying Jupyter notebook, “Capstone_Supply_Chain_Demand_Forecasting.ipynb”, submitted alongside this report in the same project folder. The notebook runs end-to-end from raw data loading through final model evaluation and is organized with Markdown headings that mirror the section structure of this report.